Marine Environmental Monitoring & Coastal Ecosystem-Preservation
Literature Review

Paper 1: Towards the Spectral Mapping of Plastic Debris on Beaches

Jenna A. Guffogg, Mariela Soto-Berelov, Simon D. Jones, Chris J. Bellman, Jennifer L. Lavers, Andrew K. Skidmore
Remote Sensing, 2021

Objective, Methodology & Key Contributions

This paper set out to test whether spectral absorption features already known to reveal plastic in laboratory settings could still be used to detect real marine plastic debris sitting on a beach among sand, seaweed and driftwood. The authors walked two beaches on the remote Cocos (Keeling) Islands with a handheld field spectroradiometer, measuring reflectance from 1 m² plots with plastic cover ranging from under 1% to about 8%, then used continuum removal and statistical testing to see which of six candidate wavelengths actually held up. Their key contribution is showing that two features, around 1215 nm and 1732 nm, remain usable even in this messy, natural background, while confirming that the relationship between how much plastic is present and how strong the signal is remains inconsistent and far from a simple rule.

Observations & Connection to My Research Interest

What stands out to me here is how honestly the paper reports its own limits - only 4 of 18 plastic-covered plots were statistically distinguishable from clean sand, and the authors don't try to oversell the result. That kind of field-level ground-truthing feels essential for my interest in satellite-based debris detection, because it's a reminder that a spectral feature proven in the lab doesn't automatically transfer to a cluttered, real-world scene. It also raises a practical question I'd like to explore further: whether combining this kind of proximal spectroscopy with drone or satellite imagery could help calibrate detection thresholds for different beach substrates.

Paper 2: Influence of Near-Surface Currents on the Global Dispersal of Marine Microplastic

David Wichmann, Philippe Delandmeter, Erik van Sebille
Journal of Geophysical Research: Oceans, 2019

Objective, Methodology & Key Contributions

The objective of this study was to check how much of a difference it makes when ocean models track microplastic not just at the surface, but slightly below it, where currents behave quite differently. Using the Parcels particle-tracking framework driven by ten years of NEMO ocean model data, the authors simulated roughly a million virtual particles under four scenarios - fixed depths, random vertical mixing, wind-driven mixing, and full 3D transport - and compared where they ended up. The main contribution is showing that the well-known subtropical 'garbage patches' weaken and disappear by around 60 m depth, while subsurface currents actively carry plastic from the subtropics toward the Arctic and Southern Ocean, giving a plausible physical explanation for plastic turning up in such remote polar waters.

Observations & Connection to My Research Interest

This paper reframes a question I'd mostly thought about in terms of where plastic is visible from space, and pushes it toward where plastic actually goes once it's no longer visible. That distinction matters a lot for remote sensing work, since satellites can only ever see the surface layer, and this study is a useful reminder that a 'clean' patch of ocean in an image doesn't mean the water underneath is clean too. It connects to my interest in monitoring plastic pollution by highlighting a blind spot in optical detection methods - one that modelling, rather than imagery, is currently better positioned to address.

Paper 3: The Physical Oceanography of the Transport of Floating Marine Debris

Erik van Sebille et al.
Environmental Research Letters, 2020

Objective, Methodology & Key Contributions

Rather than presenting new experimental results, this review's objective was to bring together, in one place, everything currently understood about the physical processes that move floating plastic around the ocean, from basin-scale gyres and eddies down to Langmuir circulation and individual breaking waves on a beach. Written by around forty oceanographers under the SCOR Working Group, the methodology was a structured literature synthesis drawing on drifter data, lab experiments, remote sensing and both Eulerian and Lagrangian ocean models. Its key contribution is organising this scattered knowledge by spatial scale and clearly identifying open questions, such as the unresolved 'missing plastic' problem, where far more plastic is estimated to enter the ocean than is ever observed floating.

Observations & Connection to My Research Interest

As a starting point for understanding the field, this review was genuinely useful - it works almost like a map of the whole research area, showing which processes are well understood and which are still guesswork. For my own interest in remote sensing of marine debris, the most relevant part is the repeated point that satellites only observe a thin, easily disturbed surface layer, while most of the interesting physics (mixing, sinking, aggregation) happens just below what any sensor can see. It made me more aware that any detection method I work with should be read alongside, rather than instead of, this kind of physical oceanographic context.

Paper 4: MARIDA: A Benchmark for Marine Debris Detection from Sentinel-2 Remote Sensing Data

Katerina Kikaki, Ioannis Kakogeorgiou, Paraskevi Mikeli, Dionysios E. Raitsos, Konstantinos Karantzalos
PLOS ONE, 2022

Objective, Methodology & Key Contributions

The goal of this paper was to build an open, realistic benchmark dataset that could train and fairly evaluate machine learning models for detecting marine debris in free Sentinel-2 satellite imagery, while also separating true debris from easily confused features like seaweed, ships, foam and turbid water. The authors gathered about 1,400 labelled image patches from 63 satellite scenes across 11 countries, using three independent annotators with a formal agreement protocol, and then tested baseline models including Random Forest and U-Net. Their key contribution is not just the dataset itself, which fills a real gap in openly available marine debris imagery, but also the finding that certain natural materials, especially driftwood and vegetation, are still very hard for any model to reliably tell apart from plastic.

Observations & Connection to My Research Interest

This paper feels the most directly applicable to hands-on remote sensing work, since it moves past showing that detection is possible in principle and instead focuses on building the reusable infrastructure (labelled data, baseline scores) that future studies need. The honesty about where the models still fail, particularly with organic material, is a useful benchmark for judging how far this technology really is from operational use. It connects well with my interest in applying deep learning to environmental monitoring, and makes me want to explore whether adding more spectral bands or higher-resolution imagery could help close that gap between debris and its natural look-alikes.

Paper 5: Finding Plastic Patches in Coastal Waters using Optical Satellite Data

Lauren Biermann, Daniel Clewley, Victor Martinez-Vicente, Konstantinos Topouzelis
Scientific Reports, 2020

Objective, Methodology & Key Contributions

This study's objective was to test whether floating plastic patches could be reliably spotted in Sentinel-2 satellite images and automatically told apart from natural floating material such as seaweed, foam and pumice. To do this, the authors developed a new spectral index, the Floating Debris Index, built spectral 'fingerprints' from both deliberately deployed plastic targets and a real, uncontrolled pollution event in Durban Harbour, and used a Naïve Bayes classifier to sort detections across four different coastlines. The key contribution is demonstrating, with real validation data rather than only simulated plastic, that this approach correctly identifies suspected plastic around 86% of the time, and it introduced an index that later studies, including MARIDA, went on to adopt.

Observations & Connection to My Research Interest

What I find most convincing about this paper is that it validates its method against a genuine pollution event, not just a tidy, staged experiment, which gives the results more real-world credibility than many earlier attempts. It also shows how a fairly simple classifier, when paired with a well-designed spectral index, can already achieve decent accuracy without needing a large deep learning model. That's encouraging for my own interest in this area, since it suggests there's still room to build lightweight, interpretable detection tools rather than always reaching for more complex architectures, especially for regions where labelled training data remains scarce.